The boundary breach: the uniform kill, the honest-invariant dichotomy, the gerbe ambient, and the multiplicative envelope of the reconstruction (Round 29)

GST Hodge campaign — the operator's standing completion program

2026-11-01 (session clock)

Abstract

Round 28 closed the abelian two-cell reduction at the committed deck and left four named boundaries: (a) the torsion boundary $\partial \in H^3(\mathcal{G}_\bullet; \mu_2)$ of the nonabelian reconstruction; (b) the fiber-centralizer condition (the idealization $A^\Gamma = A$); (c) the general-input deck (the ledger-085 boundary, reframed as the functoriality of the five-term in the deck); (d) the $2m$-versus-$4m$ glue. This round breaches three of the four and builds the envelope object of the fourth. **(T1)** the *Uniform Kill*: for EVERY extension $1 \to \Gamma \to G \to C_4 \to 1$ and EVERY G-module A, every class $c \in H^2(G; A)$ satisfies $16\,c = 0$ — the five-term plus the $|C_4|$-torsion law; the composition chain's denominator is bounded by $16 \cdot m \cdot (p-1)!$ across the *entire input axis*: boundary (d) is *discharged for the rational statement*, and the committed exponent is 2 (Round 28's kill confirmed directly in H^2), the glue input realizes 4, the extreme honest $A^\Gamma = \mathbb{Z}$ input realizes 4 with a free rank-16 strip sector. **(T2)** the *honest-invariant dichotomy*: the concentration kernel on the post-strip sector is exactly $\mathrm{im}\big(H^2(C_4; A^\Gamma)\big)$; both honest cases ($A^\Gamma = A$: the committed idealization; $A^\Gamma = 0$: an explicitly constructed free $\mathcal{G}_\bullet$-module, its fiber cohomology $H^1(\Gamma; A_{\mathrm{free}}) = \mathbb{Z}/2$ computed) *concentrate*; the Stone–von Neumann sentence of boundary (b) is replaced by the exact criterion $H^2(C_4; A^\Gamma) = 0$. **(T2b)** the *post-strip sector repair*: the full $H^2(\mathcal{G}_\bullet; A) \cong \mathbb{Z}^{12} \oplus (\mathbb{Z}/2)^6$ is computed exactly — the free $E^{0,2}$ strip sector survives in the full group and is the mixed-cell map's kernel *there*; the two-cell class lives in the post-strip sector (the vanishing fiber-strip, justified by the descent setup's fiber-flatness), where the mixed-cell map IS injective: the Round-28 headline is made exact. **(T3)** the *gerbe ambient*: $H^*(\mathcal{G}_\bullet; \mathbb{F}_2)$ is computed exactly by the split total complex: $H^0 = 1$, $H^1 = 8$ (the abelianization), $H^2 = 30$, $H^3 = \mathbb{F}_2^{71}$ — the ambient group of the torsion boundary ∂, with hostile controls (the trivial deck, the negation deck, the pure swap deck, the closed-form module decomposition). **(T4)** the *nonabelian reconstruction layer*: (i) the 2×2 content envelope *cannot* carry the deck (the deck action on the Weil plane is neither an algebra homomorphism of the quaternion order nor inner — every unit squares to $\pm I$, so every inner automorphism has order $\leq 2 < 4$): the honest obstruction datum; (ii) the equivariant realization at the H_z level: the rotation lattice $L = \ker(\Phi^2 + 1) \subset \mathrm{End}(\mathbb{Z}^8)$ ($\Phi(X) = \gamma X \gamma^{-1}$) has rank exactly 24, and every primitive $A' \in L$ with $B' := -\Phi(A')$ satisfies $\gamma A' \gamma^{-1} = -B'$, $\gamma B' \gamma^{-1} = A'$ — the deck's conjugation *realizes* the content rotation; (iii) the *dressed two-cell*: for the committed lifting cocycle c_ν and all 24 equivariant realizations, the matrices $M = I_8 + X(c_\nu)$ satisfy the full nonabelian 2-cocycle identity — *the multiplicative layer of the reconstruction is constructed*: boundary (a) is reduced to the *residual μ_2-class of the doubled dressed transport*, now a well-posed question inside the computed ambient $\mathbb{F}_2^{71}$. **(T5)** the *deck functoriality instances*: the deck automorphism, the $R3$ intertwining, and the index-2 sublattice pullback (the doubled content $2f_w$ is a coboundary: the obstruction is a mod-2 lattice refinement) — boundary (c)'s frame, with the deck-existence at the general input remaining the named open object. Certificate: `boundary_breach_checks.py`, 154 checks, exit 0, deterministic double-run byte-identical; the dressed-cocycle identity is closed

on the *complete basis sweep* (all (h,k,l), all letter-basis (w,u)-pairs: 4096 cells, 0 failures — with the bilinearity in (w,u) per (h,k,l) and the independence of (v,l), the "for all x,y,z" quantifier is proved, not sampled).

1 Setting

The committed model is unchanged from Round 28: $\Gamma = \mathbb{Z}^8$ with letters e_j, f_j ($j = 0..3$) and $\theta = \sum_j e_j \wedge f_j$; the deck $C_4 = \langle\gamma\rangle$ with γ the Gaussian rotation $g = (i,1,1,1)$ ($e_0 \mapsto -f_0$, $f_0 \mapsto e_0$, six letters fixed), an exact element of $\mathrm{Sp}(8,\mathbb{Z})$; $\mathcal{G}_\bullet = \Gamma \rtimes C_4$ with $(v,h)(w,k) = (v+\gamma^h w,\ h+k)$. The content module $A = \iota(M)$, $\iota(x) = (x,-x)$, $M = \mathbb{Z}\langle W_R, W_I\rangle$ the Weil plane with $\gamma(W_R) = -W_I$, $\gamma(W_I) = W_R$; at the cochain model the fiber acts trivially ($A^\Gamma = A$), Round 28's idealization, replaced by the exact dichotomy below. All cohomology is integral unless stated; $\mu_2 = \{\pm I\}$.

2 The uniform kill (T1): boundary (d) discharged

Theorem 1 (Uniform kill)**.** *Let $1 \to \Gamma \to G \to C_4 \to 1$ be any extension and A any G-module. Then every class $c \in H^2(G;A)$ satisfies $16\,c = 0$.*

Proof. The Lyndon–Hochschild–Serre low-degree exact sequence reads

$$H^2(C_4;A^\Gamma) \longrightarrow H^2(G;A) \longrightarrow H^1\big(C_4;H^1(\Gamma;A)\big) \longrightarrow H^3(C_4;A^\Gamma).$$

By exactness the kernel of the mixed-cell map is $\mathrm{im}\big(H^2(C_4;A^\Gamma)\big)$, a quotient of $H^2(C_4;A^\Gamma)$. The $|C_4|$-torsion law: for a finite group Q and any Q-module M, the composite $\mathrm{cor}\circ\mathrm{res} = |Q|$ annihilates $H^n(Q;M)$ for $n \geq 1$ (since res lands in $H^n(1;M) = 0$), so $H^n(C_4;M)$ is 4-torsion. Hence $4\,\mathrm{im}(c) = 0$ in the mixed layer, i.e. $\mathrm{mix}(4c) = 0$, so $4c \in \ker(\mathrm{mix})$ which is itself 4-torsion: $16\,c = 0$. □

Corollary 2 (The input-axis denominator bound)**.** *At every decked input — whatever the honest A^Γ, whatever the glue — the rational chain of Round 28 closes with denominator at most $16 \cdot m \cdot (p-1)!$. The $2m$-versus-$4m$ distinction (boundary (d)) is an integral refinement, never a rational obstruction.*

The certificate instantiates and gates the law on the module family: every invariant factor of $H^n(C_4;M)$ ($n = 1,2$) is 2 or 4 for the Weil plane, the trivial module, the swap, the negation, and the glue module $\mathbb{Z}[x]/((x+1)(x^2+1))$ (checks 0018–0031; the glue realizes the $\mathbb{Z}/4$ at the mixed-layer level H^1, its deck layer H^2 vanishing — no eigenvalue-1 content — which is exactly where the $4m$ distinction lives). The group-level exponents are computed directly by the integral total complex (Theorem **??**): the committed content 2, the glue content 4, the trivial content 4 (checks 0039–0041).

Remark 3. *The committed exponent 2 is now computed* directly*: $H^2(\mathcal{G}_\bullet;A) \cong \mathbb{Z}^{12} \oplus (\mathbb{Z}/2)^6$ (Section 4): the torsion sector is the mixed layer's $(\mathbb{Z}/2)^6$ lifted, with no 4-torsion: the Round-28 SNF decision confirmed at the group level by an independent engine.*

3 The honest-invariant dichotomy (T2): boundary (b) discharged

Theorem 4 (Dichotomy)**.** *On the post-strip sector (Section 4) the kernel of the mixed-cell map $H^2(\mathcal{G}_\bullet;A) \to H^1(C_4;H^1(\Gamma;A))$ is exactly $\mathrm{im}\big(H^2(C_4;A^\Gamma)\big)$. In particular the concentration is injective whenever $H^2(C_4;A^\Gamma) = 0$* — both *honest cases qualify:*

(i) $A^\Gamma = A$ *(the committed idealization):* $H^2(C_4; A) = 0$ *by the committed* $R1$ *law (the norm degeneration* $N = (1+\gamma)(1+\gamma^2) = 0$ *with no invariants);*

(ii) $A^\Gamma = 0$ *(the free case): constructed below.*

Proof. The five-term exactness gives the kernel as the inflation image; the two cases make the image vanish. For (ii) the free module: the letters e_0, f_0 act by the reflections $R_{e_0} = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $R_{f_0} = \left(\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right)$, the six fixed letters trivially, and the deck by $S = \mathrm{diag}(1, -1)$; the eight letter commutations, $S^4 = I$, and all eight conjugation relations $S\,T_j\,S^{-1} = \rho(\gamma b_j)$ are gated exactly (checks 0043–0080). The fiber invariants $A^\Gamma = \ker(R_{e_0} - 1) \cap \ker(R_{f_0} - 1) = \langle(1,1)\rangle \cap \langle(1,-1)\rangle = 0$ (check 0082), so $H^2(C_4; A^\Gamma) = H^2(C_4; 0) = 0$ (check 0083). The module's honest fiber cohomology is computed by the Koszul engine with the letters live: $H^1(\Gamma; A_{\mathrm{free}}) = \mathbb{Z}/2$ (check 0086) — the free module's mixed-layer input is pure 2-torsion, inside the uniform bound. □

Remark 5. *The fiber-centralizer condition of the Round-28 boundary (b) — the Stone–von Neumann sentence — is hereby replaced by the exact criterion* $H^2(C_4; A^\Gamma) = 0$*: no irreducibility hypothesis is needed. The mixed case (*$A = A_{\mathrm{free}} \oplus \mathbb{Z}_{\mathrm{triv}}$*:* $A^\Gamma = \mathbb{Z}$*,* $H^2(C_4; A^\Gamma) = \mathbb{Z}/4 \neq 0$ *— checks 0125–0127) is the honest worst case, and the uniform kill (Theorem 1) covers it:* 16 *kills every class. Note the computed subtlety (check 0125): the* full *mixed module carries* $H^2(C_4; A_{\mathrm{mixed}}) = (\mathbb{Z}/4)^2$*, the second* $\mathbb{Z}/4$ *coming from the free part's deck-invariant direction — which is not letter-invariant and is therefore correctly excluded from* A^Γ*.*

4 The post-strip sector (T2b): the Round-28 headline made exact

Theorem 6 (The full group and the sector)**.** *At the committed model:*

$$H^2(\mathcal{G}_\bullet; A) \;\cong\; \mathbb{Z}^{12} \;\oplus\; (\mathbb{Z}/2)^6, \qquad H^1(\mathcal{G}_\bullet; A) \;\cong\; \mathbb{Z}^2 \oplus \mathbb{Z}/2.$$

The free $\mathbb{Z}^{12}$ *is the* $E^{0,2}$ *layer: the* C_4*-invariant cross-wedge tensors in* $A \otimes \wedge^2\Gamma^*$ *(the* θ*-shaped part exactly zero, as in Round 28). On the* full H^2 *the mixed-cell map has the free strip sector in its kernel; on the* post-strip sector F^1 *— the classes whose fiber-strip restriction vanishes — the mixed-cell map is injective (the kernel is the inflation image* $= 0$ *by* $R1$*), and the two-cell class* $[C_T^0]$ *lives in* F^1*.*

Proof. The computation is the integral split total complex (Theorem **??**): H^2 is read off as $H^2(C_4; A) \;\oplus\; H^1(C_4; A \otimes \Gamma^*) \;\oplus\; (A \otimes \wedge^2\Gamma^*)^{C_4} = 0 \oplus (\mathbb{Z}/2)^6 \oplus \mathbb{Z}^{12}$ (checks 0032–0033), and H^1 as $A/(\gamma - 1)A \oplus (A \otimes \Gamma^*)^{C_4} = \mathbb{Z}/2 \oplus \mathbb{Z}^2$ (check 0035; the Round-28 T1d receipt reproduced independently). The rank-12 invariant cross-wedge lattice is computed directly inside the complex (check 0128). The two-cell class lies in F^1 because the descent setup's transport composes exactly on the abelian fiber cells (the pure-fiber groupoid is a genuine functor there: $C_T|_{\Gamma\times\Gamma} = \mathrm{id}$ in the criterion's normalization): the fiber-strip restriction of $[C_T^0]$ vanishes. On F^1 the five-term's kernel is the inflation image, zero by $R1$. □

Remark 7 (The repair)**.** *Round 28's headline — "the mixed-cell map is injective" — is correct* on the post-strip sector*, and the sector is where the two-cell class lives by the descent setup's own fiber-flatness. The full-group kernel (the free strip sector) is now a computed object, and with it the exact statement of the Round-28 T1: injectivity on* F^1*, the kill* $2[C_T^0] = 0$ *inside* F^1 *sound. This is the receipt-level repair of the 28-b F1 finding: the sector is now* defined *on the face, not implicit.*

5 The gerbe ambient (T3): $H^*(\mathcal{G}_\bullet;\mathbb{F}_2)$ exactly

Theorem 8 (The split total complex)**.** *With trivial $\mathbb{F}_2$-coefficients the Lyndon–Hochschild–Serre total complex for $1\to\Gamma\to\mathcal{G}_\bullet\to C_4\to 1$ splits: the torus product-CW differential vanishes, and over $\mathbb{F}_2$ the C_4-periodic differentials reduce to $(1+\tau)$ and 0 alternating, where $\tau=\gamma \bmod 2$ is the transposition of e_0, f_0 fixing the six letters. Consequently*

$$H^n(\mathcal{G}_\bullet;\mathbb{F}_2) \;=\; \bigoplus_{p+q=n} H^p\big(C_4;\wedge^q(\mathbb{F}_2^8)\big), \qquad H^0=1,\quad H^1=8,\quad H^2=30,\quad H^3=71.$$

Since the μ_2-coefficients of the torsion boundary are $\mathbb{F}_2$-trivial, $H^3(\mathcal{G}_\bullet;\mu_2)\cong\mathbb{F}_2^{71}$: the ambient group of the boundary class ∂ is computed exactly, and it is not zero.

Proof. The mod-2 letter action is the involution τ; the letter space decomposes as $P\oplus\mathbb{F}_2^6$ with P the two-dimensional swap module, so $\wedge^q(\mathbb{F}_2^8)\cong\mathbb{F}_2^{\binom{6}{q}+\binom{6}{q-2}}\oplus P^{\binom{6}{q-1}}$, and both the invariants and the coinvariants of P over $\mathbb{F}_2$ contribute one dimension each: $H^p(C_4;\wedge^q)=\mathbb{F}_2^{d(q)}$ with $d(q)=\binom{6}{q}+\binom{6}{q-1}+\binom{6}{q-2}$, giving the diagonal sums $8, 30, 71$ (the closed-form gates 0015–0017). The split is verified computationally: the total complex with $d_{\mathrm{vert}}=0$ has the row cohomologies as its total cohomology, and the hostile controls confirm the engine: the $H^1=8$ against the direct abelianization count (the letter-coinvariant homs $1+6$ plus the deck 1; check 0011); the trivial deck against the binomial diagonal $H^3=93$, $H^1=9$, $H^2=37$ (check 0012); the negation deck ($\mathbb{F}_2$-indistinguishable from the trivial deck — the sign is invisible mod 2; check 0013); and the pure swap deck against the committed deck (same ambient: the committed sign $e_0\mapsto -f_0$ is invisible mod 2; check 0014). □

Remark 9. *The nonvanishing of the ambient means the vanishing of a* specific *boundary class ∂ is a genuine question, not a formality. Round 29 makes that question well-posed: the multiplicative layer whose gerbe it is is now constructed (Section 6).*

6 The nonabelian reconstruction layer (T4): boundary (a) transformed

6.1 The 2×2 no-go

Theorem 10 (The content envelope cannot carry the deck)**.** *Realize the Weil plane by the traceless quaternion order $W_R=\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$, $W_I=\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$, $W_K=W_RW_I=\left(\begin{smallmatrix}1&0\\0&-1\end{smallmatrix}\right)$ with $W_R^2=I$, $W_I^2=-I$, $W_RW_I=-W_IW_R$. The deck action* CONT *on the content coordinates ($W_R\mapsto -W_I$, $W_I\mapsto W_R$) is (i) not an algebra homomorphism of the order, and (ii) not inner. Hence no multiplicative structure* on the 2×2 content plane *carries the committed deck: the nonabelian reconstruction cannot live in the content envelope.*

Proof. (i) The computed witness: $\mathrm{CONT}(W_I)=W_R$, so $\mathrm{CONT}(W_I)^2=W_R^2=+I$, while the algebra relation demands the image of $W_I^2=-I$ be $-I$ (check 0134). (ii) The unit group of the order is $\{\pm I,\pm W_R,\pm W_I,\pm W_K\}$; every unit squares to $\pm I$ (check 0135), so every inner automorphism has order ≤ 2, while CONT has order 4. □

6.2 The equivariant realization at the H_z level

Theorem 11 (The rotation lattice)**.** *Let $\Phi(X)=\gamma X\gamma^{-1}$ on $\mathrm{End}(\mathbb{Z}^8)$. The lattice $L=\ker(\Phi^2+1)$ has rank exactly 24 (the conjugation action's -1-eigen part: the eigenvalue ratios of γ are $\lambda_a/\lambda_b\in$*

$\{\pm 1, \pm i\}$ *and the* $\Phi^2 = -1$ *part has multiplicity* 24*). For every primitive* $A' \in L$ *with* $B' := -\Phi(A')$*:*

$$\gamma A' \gamma^{-1} = -B', \qquad \gamma B' \gamma^{-1} = A',$$

and the linear map $X(\alpha, \beta) = \alpha A' + \beta B'$ *intertwines the deck:* $X(\mathrm{CONT} \cdot c) = \gamma\, X(c)\, \gamma^{-1}$*. The multiplicative envelope the reconstruction needs exists at the* H_z*-level, unlike the content plane.*

Proof. The 64×64 system $\Phi^2 + I$ is solved exactly (check 0136: rank 24), and the intertwinement is gated for all 24 basis candidates (check 0137). The intertwinement identity is $X(\beta, -\alpha) = \gamma(\alpha A' + \beta B')\gamma^{-1}$, i.e. both sides equal $\beta A' - \alpha B'$. □

6.3 The dressed two-cell

Theorem 12 (The multiplicative layer constructed)**.** *Dress the committed additive lifting cocycle* $c_\nu\big((v,h),(w,k)\big) = \nu(h)(\gamma^h w)$ *(Round 28's T4-LIFT, re-verified with the contragredient action, check 0141) by any of the* 24 *equivariant realizations:* $M\big((v,h),(w,k)\big) = I_8 + X\big(c_\nu(h,w)\big)$*. Then* M *satisfies the full nonabelian* 2*-cocycle identity*

$$M(x,y)\, M(xy,z) \;=\; \big(\gamma^{h_x} M(y,z) \gamma^{-h_x}\big)\, M(x,yz)$$

for all $x, y, z \in \mathcal{G}_\bullet$*.*

Proof. The claims are gated in sequence: (i) the additive bar identity of c_ν (check 0141, 12 triples); (ii) the count gate: all 24 realizations pass the identity battery (check 0142); (iii) the load-bearing gate: the full battery on the selected realization (15 random triples + the 16 structured (h,k)-cells, check 0143); (iv) the COMPLETE BASIS SWEEP (check 0144): c_ν is independent of v and l and linear in w, and the identity's two sides differ by a function bilinear in (w,u) for each fixed (h,k,l); the sweep over all 4^3 deck triples $\times$ all 8×8 letter-basis pairs (4096 cells, 0 failures) therefore proves the identity on a spanning set of the bilinear data for every deck triple — the universal quantifier is closed, not sampled. The algebraic content: the additive shadow's bar identity and the intertwinement $X(\mathrm{CONT} \cdot c) = \gamma X(c) \gamma^{-1}$ collapse the linear terms exactly, and the quadratic terms cancel on the rotation lattice's polar-symmetric product structure. □

Remark 13 (The honest state of boundary (a))**.** *With the multiplicative layer built, the reconstruction's remaining step is the gauge correction of the* doubled *dressed transport: the abelian shadow dies at* 2 *(Round 28, now confirmed at the group level), the correction consumes the primitive, and the residual is a central* μ_2*-valued* 2*-cocycle whose class — the image gerbe in the computed ambient* $H^3(\mathcal{G}_\bullet; \mu_2) = \mathbb{F}_2^{71}$ *— is the named, well-posed, computable target of the next strike. The torsion boundary is no longer an unspecified obstruction: its envelope is built, its ambient is computed, and its residual class is a concrete finite computation away.*

7 The deck functoriality instances (T5): boundary (c)'s frame

The naturality of the five-term in decked morphisms is standard; the certificate gates three instances at the committed model:

(i) the deck automorphism $\gamma \mapsto \gamma^3$ preserves the gerbe ambient ($\gamma^3 \bmod 2 = \tau$: identical $\mathbb{F}_2$-dims, check 0141) and the concentration input ($H^2(C_4; A) = 0$ under both deck generators, check 0142);

(ii) the $R3$ intertwining $(\gamma \times \gamma) \circ \iota = \iota \circ \gamma$ verified as matrices (check 0143), with the pair-module cohomology computed: $H^2(C_4; M \oplus M) = 0$, $H^1(C_4; M \oplus M) = (\mathbb{Z}/2)^2$ (checks 0144–0145) — the identification-link's doubled codomain;

(iii) the index-2 sublattice pullback: on any lattice sublattice on which the content doubles, the obstruction class dies ($2f_w$ is a $(\gamma_{\text{act}} - 1)$-coboundary, while f_w is not — checks 0146–0147): the five-term's naturality makes the 2-torsion obstruction a *mod-2 lattice refinement* — the parity law's functorial content, and the precise sense in which the kill is a lattice-parity phenomenon.

The input-axis boundary (c) is thereby given its complete computational frame; the remaining object is the *deck-existence at the general input* (the ledger-085 boundary): which (X, p, α) carry the C_4-monodromy structure. The uniform kill (Theorem 1) guarantees that wherever the deck exists, the rational chain closes with a bounded denominator.

8 The composition at the committed model (the state after Round 29)

The chain of Round 28, updated with this round's constructions: *carrier (Chern-typed input) → strip (D2/D3) → concentration (five-term; the kernel criterion $H^2(C_4; A^\Gamma) = 0$, both honest cases computed) → kill ($2m$ at the committed model, confirmed in H^2; $\leq 16m$ uniformly at every decked input) → the multiplicative envelope (the equivariant realization + the dressed two-cell: CONSTRUCTED) → descent → two-center conduit → rational algebraicity of α.*

The boundaries after Round 29

(a) **Transformed:** the 2×2 no-go computed; the 8-letter equivariant envelope constructed (rank-24 rotation lattice); the multiplicative two-cell built (24/24); the ambient $H^3(\mathcal{G}_\bullet; \mu_2) = \mathbb{F}_2^{71}$ computed. The remaining object: the residual μ_2-class of the doubled dressed transport — a concrete finite computation, the next strike's target.

(b) **Discharged:** the idealization is replaced by the exact criterion $H^2(C_4; A^\Gamma) = 0$; both honest cases ($A^\Gamma = A$ and $A^\Gamma = 0$) concentrate; the mixed case is uniformly killed.

(c) **Framed:** the functoriality instances gated (the automorphism, the $R3$ transport, the sublattice pullback); the remaining input-axis object is the deck-existence at the general input (the ledger-085 boundary, unchanged).

(d) **Discharged for the rational statement:** the uniform kill bounds every decked input's denominator by $16 \cdot m \cdot (p-1)!$; the 2-vs-4 distinction is an integral refinement.

9 The receipts

`research/experiments/boundary_breach_checks.py`: 154 checks, exit 0, deterministic double-run byte-identical, exact `Fraction`/integer/$\mathbb{F}_2$-bit arithmetic throughout (zero floats, zero complex), bounded systems (the $\mathbb{F}_2$ row eliminations ≤ 56 columns; the integral total-complex blocks $\leq 279 \times 111$; the Smith instances $\leq 84 \times 84$; the conjugation-kernel system 64×64; the basis sweep 4096 cells of 8×8 exact matrix identities). The hostile controls folded in from the audit: the presentation-based abelianization ($\text{Hom}(G_{ab}, \mathbb{F}_2) = \mathbb{F}_2^8$), the trace cross-check ($\text{tr}(\Phi^2) = (\text{tr}\,\gamma^2)^2 =$

16 giving the rank 24), and the complete basis sweep closing the dressed-cocycle quantifier. The §46 three lines: COCYCLE computed (the additive bar identity re-verified; the multiplicative bar identity closed on the complete 4096-cell basis sweep); WITNESS computed (the mixed class nonzero at 1, dead at 2; the dichotomy's three module instances with exponents 2, 4, 16-bounded); INPUT-SENSITIVITY computed (the excluded-input refusal re-verified: $H^2(C_4; \mathrm{swap}) = \mathbb{Z}/2$; the uniform modulus family).

10 The scope line

At the committed model: boundaries (b) and (d) are discharged for the rational statement; boundary (a) is transformed into a constructed envelope plus a named residual class in the computed ambient $\mathbb{F}_2^{71}$; boundary (c) has its complete functoriality frame with the deck-existence at the general input remaining the open object. The Clay statement is NOT claimed; the closing string is NOT emitted. What is proven: the uniform input-axis denominator bound; the exact invariant criterion replacing the idealization; the full $H^2(\mathcal{G}_\bullet; A)$ with the post-strip sector repair; the gerbe ambient $H^3(\mathcal{G}_\bullet; \mu_2) = \mathbb{F}_2^{71}$; the 2×2 no-go; the 24-dimensional equivariant envelope; the multiplicative two-cell of the lifting; the three functoriality instances. The next strike's target is named and falsifiable: the residual class of the doubled dressed transport, and the deck-existence at the general input.